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On behalf of Pirro Hysi, Felyx Wong, Manik Sharma, Alastair Denniston, Andrew Dick, Becca Edward, Simon Epps, Ed Hughes, Jen Jung, Sharon Kerr, Jess Kraker, Phil Murray, Alan Palestine, Michael Paley, Mina Pantcheva, Amit Reddy, Miles Stanford, Will Tucker, Graham Wallace, Wayne Yokoyama, and the i3eye Genomic Consortium

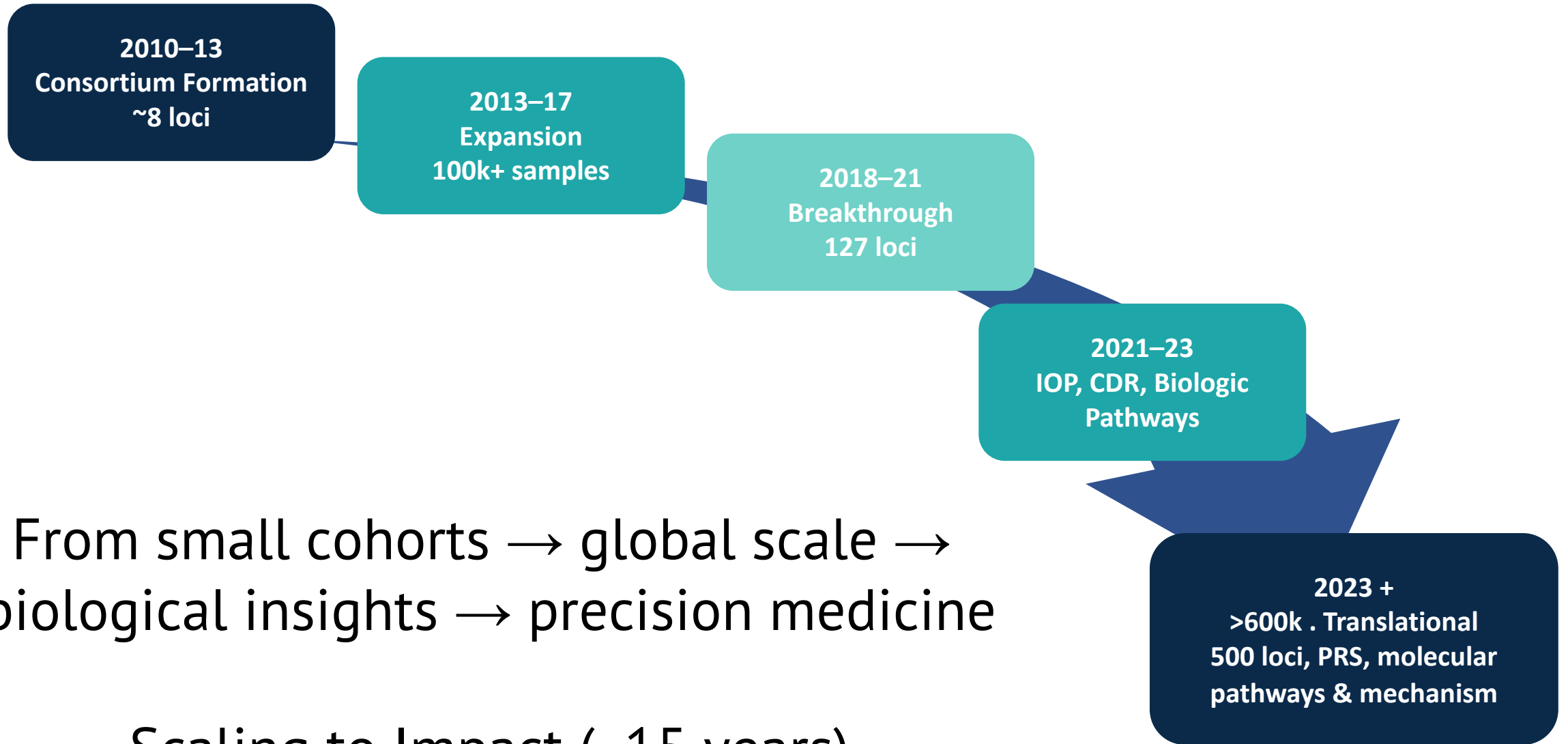
What do we know about uveitis genomics?

- **Kuiper JJ, et al. 2014.** GWAS identifies a functional ERAP2 haplotype associated with birdshot chorioretinopathy.
- **Robinson PC, et al. 2015.** Acute anterior uveitis and ankylosing spondylitis. Similarities and differences.
- **Huang XF, et al. 2020.** GWAS in AAU – new loci.
- **de Groot EL, et al. 2023.** Association of Risk Variants in the CFH Gene With Elevated Levels of Coagulation and Complement Factors in Idiopathic Multifocal Choroiditis.
- **Gelfman S, et al. 2023.** GWAS meta-analysis in anterior uveitis.
- **Koskimaki F, et al. 2025** GWAS in anterior uveitis.

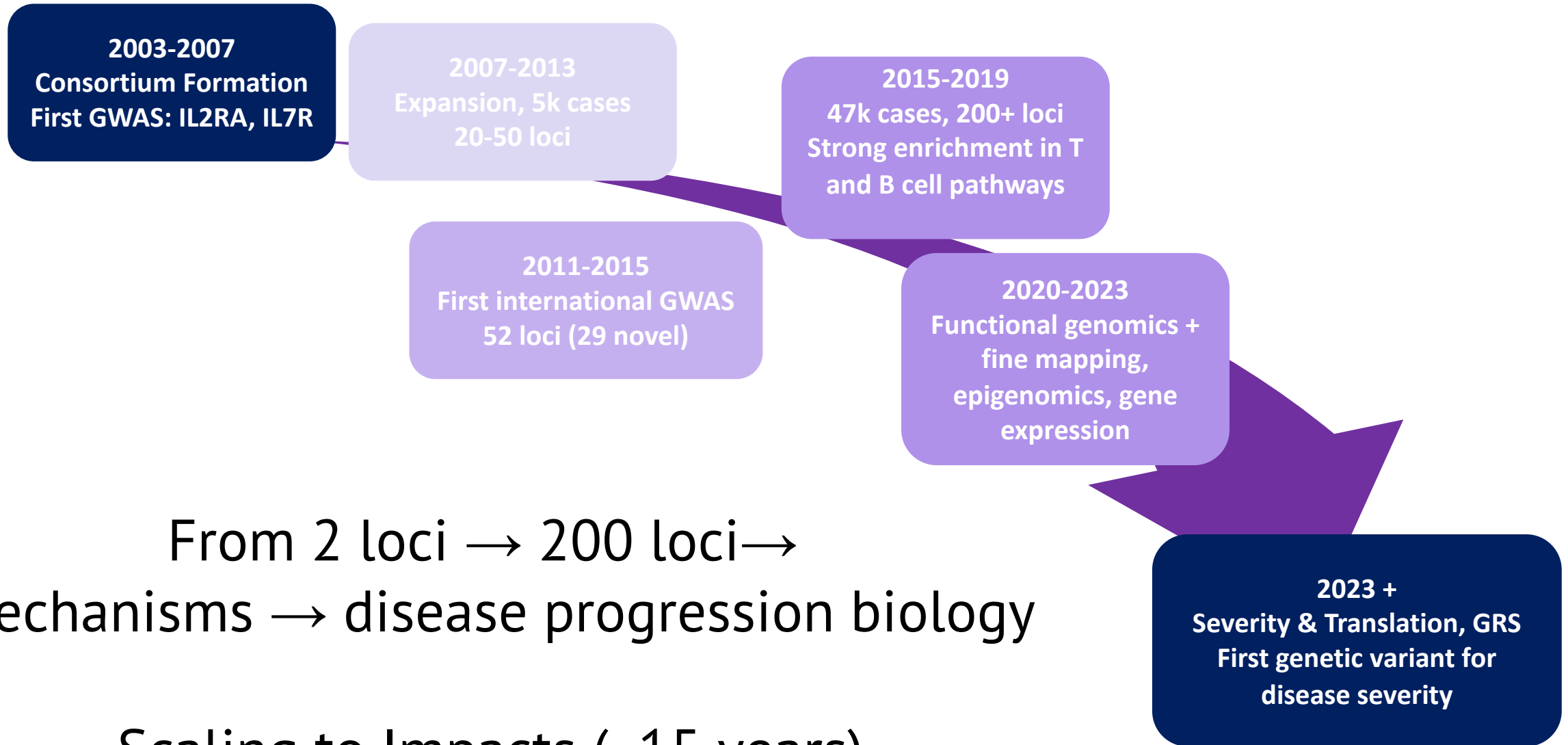
Eye rationale

- Inflammatory eye diseases are **rare, heterogeneous, and underpowered for genetics**
- Single-centre cohorts → **miss modest-effect variants**
- Biobanks
 - ICD codes lack phenotypic precision
 - Intermediate and posterior uveitis infrequent
- Need **global scale + diversity** to:
 - Detect genetic risk
 - Understand shared immune pathways (e.g. MS, sarcoidosis)
 - Enable **precision medicine and trial stratification**

International Glaucoma Genetics Consortium



International MS Genetics Consortium



From 2 loci → 200 loci →
mechanisms → disease progression biology

Scaling to Impacts (~15 years)

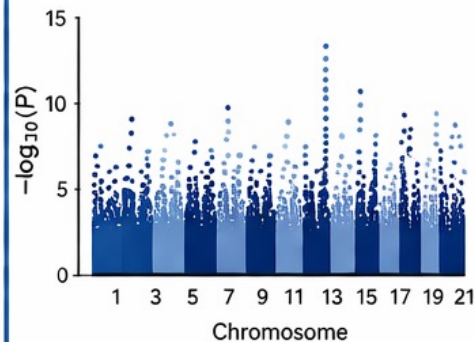
THE 4 PILLARS OF SUCCESS

Lessons from Global Genomics Consortia



1. SCALE

Large sample sizes are essential for discovery and replication.

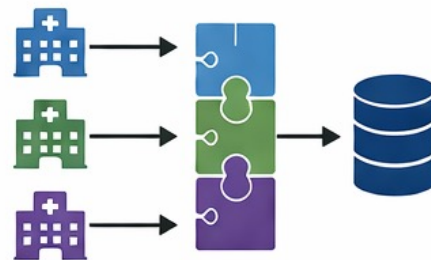


*More data → More power →
More discoveries*



2. HARMONISATION

Consistent phenotyping, data standards and analytical approaches enable meaningful meta-analysis.



*Different sites,
one standard,
shared data, stronger results*



3. DIVERSITY

Including diverse ancestries and populations improves generalizability and reveals novel biology.



*Representation matters –
for discovery, equity
and impact*



4. TRUST & GOVERNANCE

Transparent governance, data sharing and mutual respect build trust and sustain collaboration.



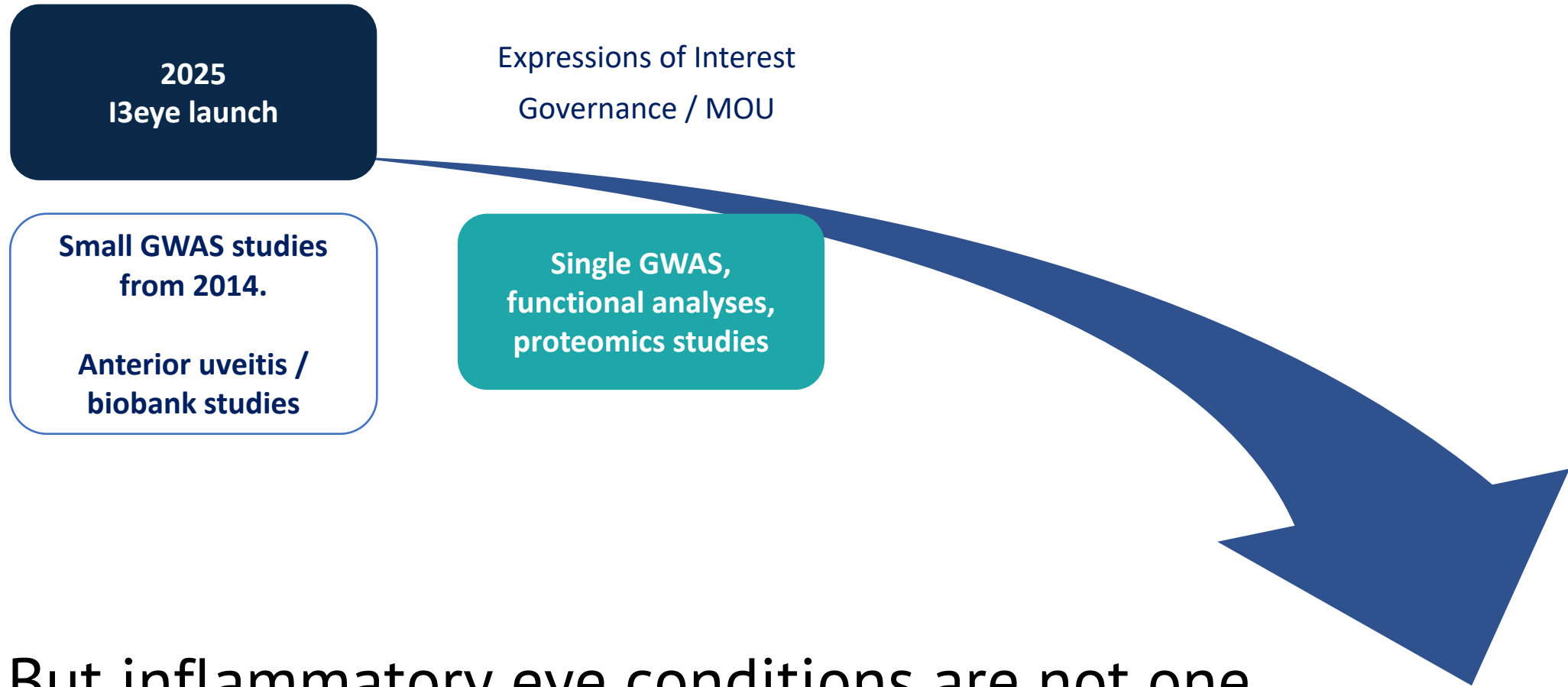
*Ethics, data security,
fair access and shared credit
drive long-term success*



Together, these pillars turn collaboration into discovery



i3eye Genomics Consortium: where are we?



But inflammatory eye conditions are not one disease

i3eye Genomics Consortium: Strategic Vision

**2025
i3eye launch**

Many centres, contributing small and well phenotyped 'n', with multiomics, over a long period of time -> Stratified meta-analyses

Expressions of Interest from both academic biobanks (Unavailable to Industry) and smaller case/control studies, and also potential new clinical academic collaborators, not yet collecting samples

Governance / MOU

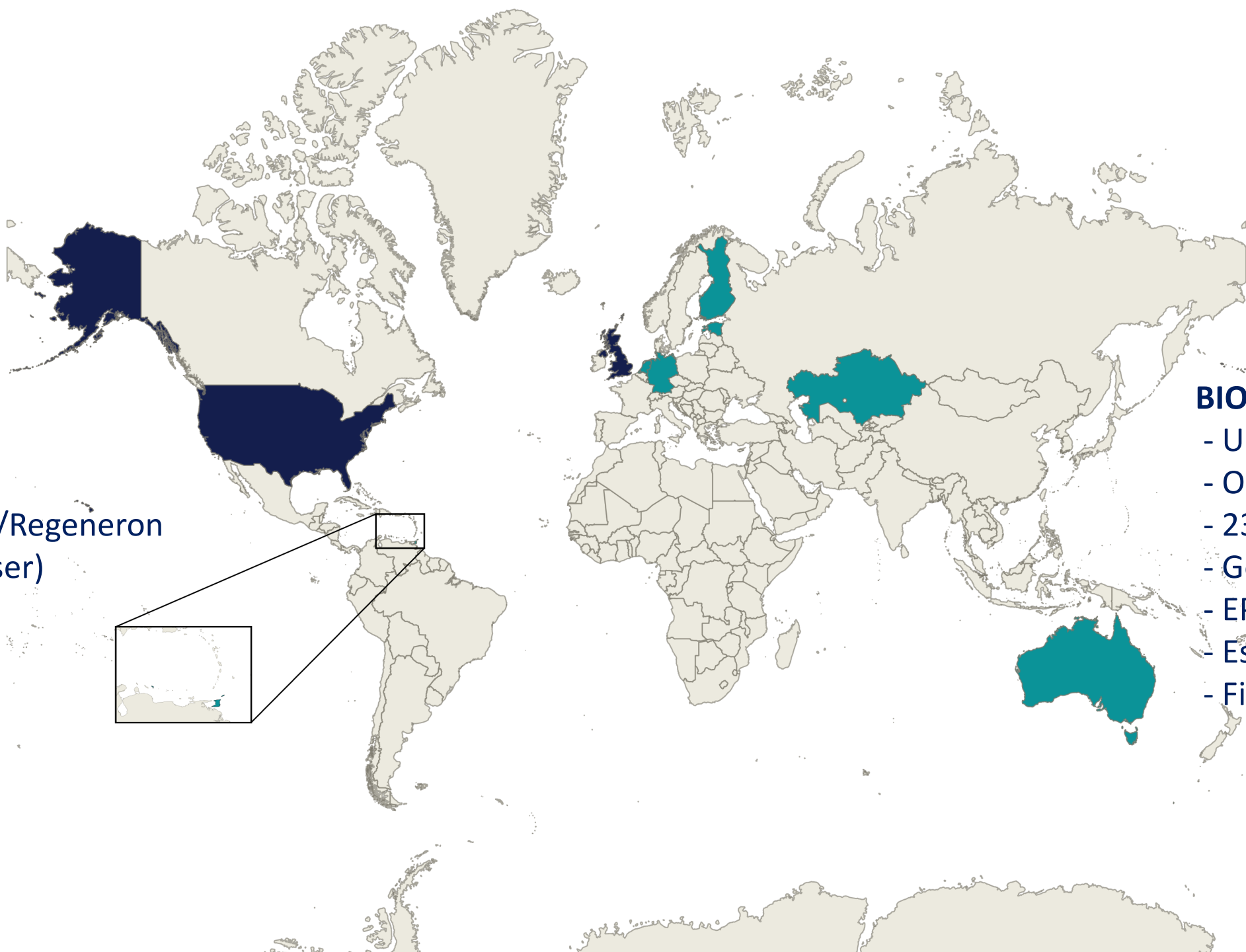
Harmonise data collection fields

Not only drug discovery, but deep phenotyping to yield clinically actionable risk stratification for clinical care management

i3eye

BIOBANKS

- MVP
- All of Us
- Geisinger/Regeneron
- JERA (Kaiser)



BIOBANKS

- UKBB
- OurFutureHealth
- 23andMe
- Genes&Health
- EPICNorfolk
- Estonian BB
- FinnGen

The history of genomics consortia tells us that rare, heterogeneous, immune-mediated eye diseases will not yield their biology to isolated studies or biobank meta-analyses alone. They require what i3Eye is building: phenotypic enrichment at scale, diversity, harmonisation and trust, for stratified meta-analyses.

Joining i3Eye: What You Give. What You Gain.

*A global genomics collaboration to transform understanding and outcomes
in inflammatory eye diseases*

WHAT YOU CONTRIBUTE



SAMPLES & DATA

Phenotyped cohorts
(uveitis, optic neuritis,
scleritis)



STANDARDISATION

Shared definitions and
protocols (SUN criteria
for uveitis, ICON criteria
for optic neuritis)



COLLABORATION

Input into study design,
data curation and
analyses



**YOUR DATA STAYS
YOURS —**
but together, we achieve
global scale.

WHAT YOU GAIN



SCALE

Access to large,
international datasets
and diverse ancestries



DISCOVERY POWER

Well-powered genetics
to detect meaningful
risk variants



AUTHORSHIP & LEADERSHIP

Lead analyses, first-author
opportunities and
pre-publication access
to results



IMPACT

Advance precision medicine,
clinical trials and novel
therapeutics



INDEPENDENT ANALYSES

Permitted with transparency;
avoid competing with
consortium analyses



FAIR AUTHORSHIP

Based on ICMJE criteria;
contribution-based,
not site-based



SECURE & ETHICAL DATA SHARING

GDPR-compliant and
consent-respecting



TRUST & COLLABORATION

Built on openness, respect,
confidentiality and
timely engagement



LOW BARRIER TO ENTRY

No mandatory financial
commitment

What i3Eye offers is simple: you keep ownership of your data, but gain the power of global scale through a long-term **academic** collaboration—within a framework that ensures fairness, credit, and scientific impact

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